

01 Background & Motivation

Inverter-based resources are rapidly replacing synchronous generators, making the safety of inverter current control increasingly critical to grid reliability.

- **IEEE 1547** limits injected current to 1.1 pu of rated value, yet conventional controllers routinely violate this during faults and voltage sags^[1].
- **PI Controllers** are simple and widely used but they overshoot during transients and rely on saturation blocks as a post-hoc fix^[2].
- **LQR** improves dynamic response but still lacks explicit current limit enforcement^[3].
- **Conventional Safety Measures**:- saturation blocks, heuristic clamps, conservative tuning are reactive and not guaranteed

02 The Gap we Fill

CLF-CBF safe control has been applied to power systems but only at the outer loop (slow voltage and frequency timescales). It is non-trivial to address the inner current control loop, where:

- Transients develop in milliseconds
- The controller must be compatible with high-frequency PWM
- Reference currents may temporarily exceed safe limits during sags, creating a direct conflict between tracking and safety

03 Study System

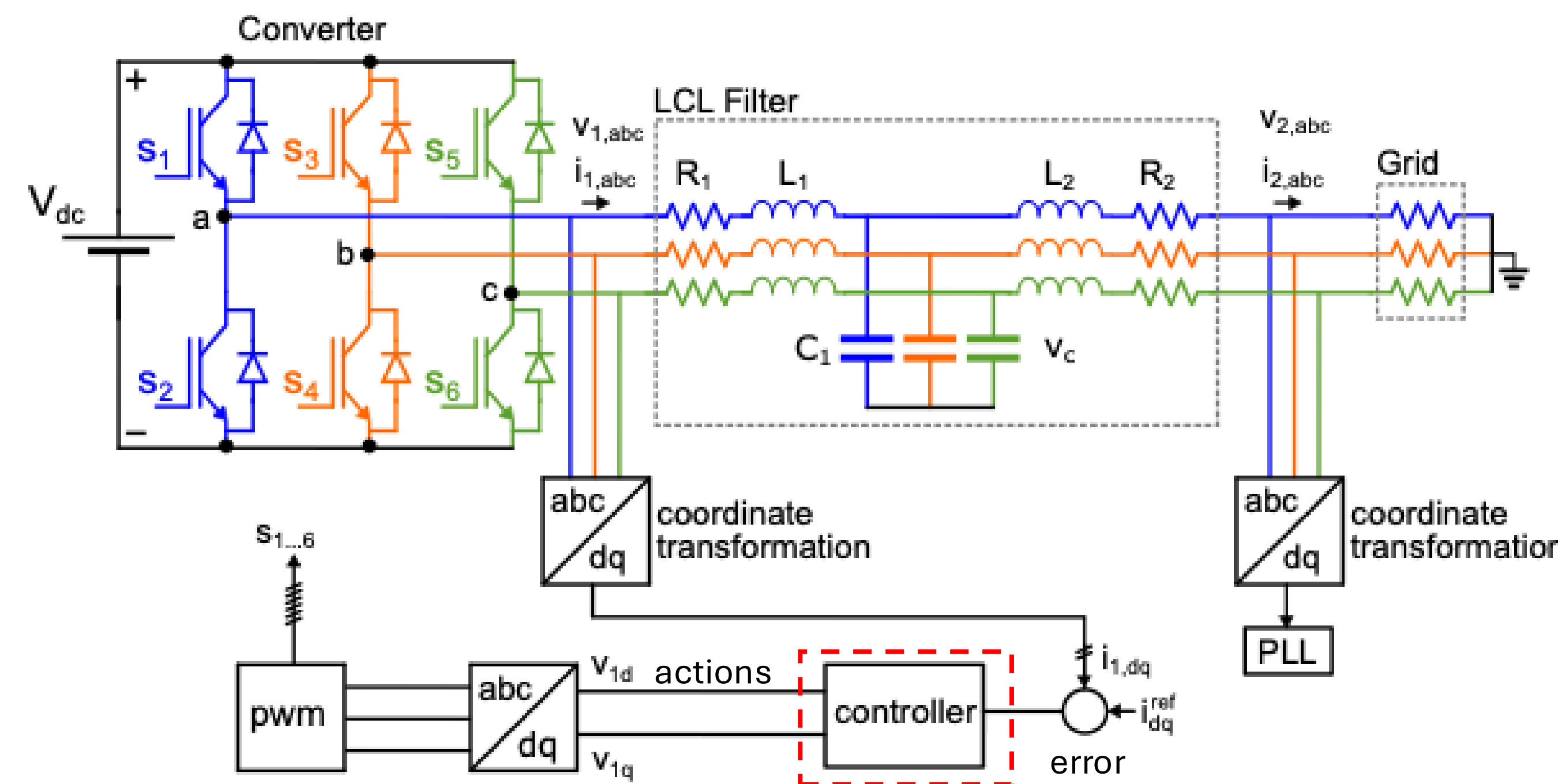


Fig. 1 – Three-phase grid-following inverter with LCL filter. State vector $x = [i_{1d}, i_{1q}, i_{2d}, i_{2q}, v_{cd}, v_{cq}]^T$. Dynamics: $\dot{x} = Ax + Bu$

SAFE CURRENT LIMIT (IEEE 1547)

$$I_{max} = \sqrt{\frac{2}{3}} \frac{1.1S}{V_{LL,rms}} = 224.5 \text{ A}$$

04 Simulation Parameters

Parameter	Value
Rated Power S	100 kVA
DC-Link Voltage V_{dc}	1500 V
Grid Voltage V_{LL}	400 V
Grid Frequency f	60 Hz
Switching Frequency f_s	10 kHz
Max d-axis Current	224.5 A
CBF Gain $\gamma_d = \gamma_q$	6000

05 Control Lyapunov Function

The CLF is built from the LQR Riccati solution, guaranteeing asymptotic convergence of inverter currents to their reference:

CLF DEFINITION

$$\dot{v}(x) = (x - x^{ref})^T P (x - x^{ref})$$

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

P is the unique positive-definite solution to the algebraic Riccati equation. Minimizing $\dot{v}(x, u)$ drives the fastest descent toward the reference

06 Control Barrier Functions

Separate CBFs enforce d- and q-axis current limits, making the safe set **forward invariant**:

BARRIER FUNCTIONS

$$h_d(x) = i_d^{max} - i_{1d} \geq 0$$

$$h_q(x) = i_q^{max} - i_{1q} \geq 0$$

FORWARD INVARIANCE CONDITION

$$\nabla_x h(x)^T \dot{x} + \alpha_h(h(x)) \geq 0$$

If this condition is satisfied for all time, the system cannot leave the safe set. Both CBFs are relative-degree-one

07 Super-linear Barrier and γ Bound

SUPER-LINEAR α_H

$$\alpha_h(z) = \gamma(z+1) \log(z+1) \leq z, f_s$$

γ BOUND

$$\gamma \leq \frac{f_s}{(h^{max}+1) \cdot \log(h^{max}+1)}$$

Smooth control far from the limit; aggressive enforcement near the boundary

08 Unified CLF-CBF-QP

QUADRATIC PROGRAM - SOLVED EVERY SWITCHING CYCLE

$$u^* = \underset{s, t}{\operatorname{argmin}} 2(x - x^{ref})^T P B u + \lambda \|u\|^2$$

$$s, t \nabla h_d(x)^T \dot{x} + \gamma_d(h_d + 1) \log(h_d + 1) \geq 0$$

$$\nabla h_q(x)^T \dot{x} + \gamma_q(h_q + 1) \log(h_q + 1) \geq 0$$

The CLF cost drives tracking; the CBF constraints are hard walls. Safety always wins. When far from the limits, the QP behaves like a pure tracker.

09 Safety: Peak Current

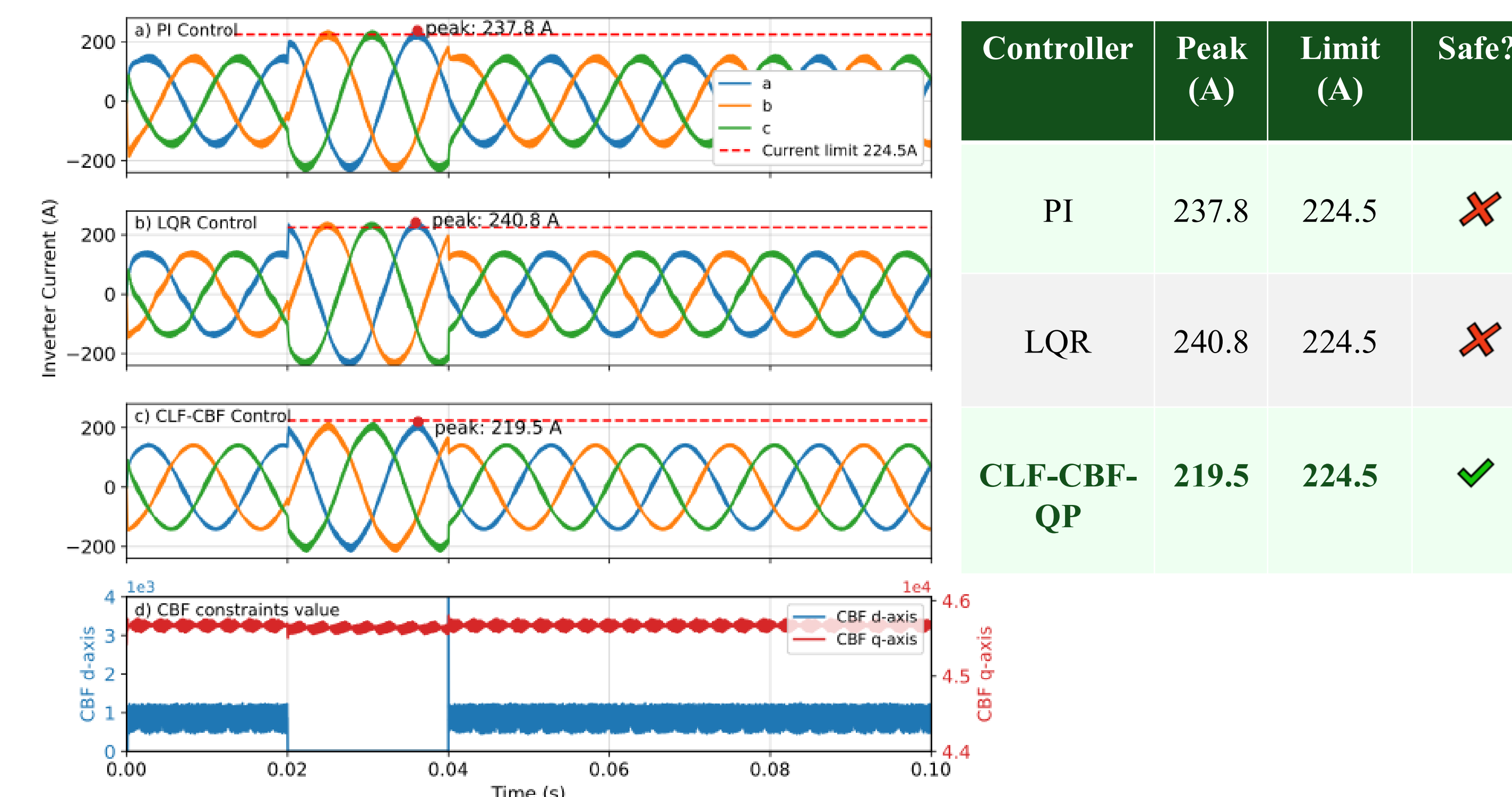


Fig. 2 – Three-phase inverter current under 0.6 pu voltage sag (0.02-0.04s). Red dashed line = 224.5 A safe limit. Bottom Panel: CBF constraint values :- strictly positive

10 Dynamics: Current Tracking

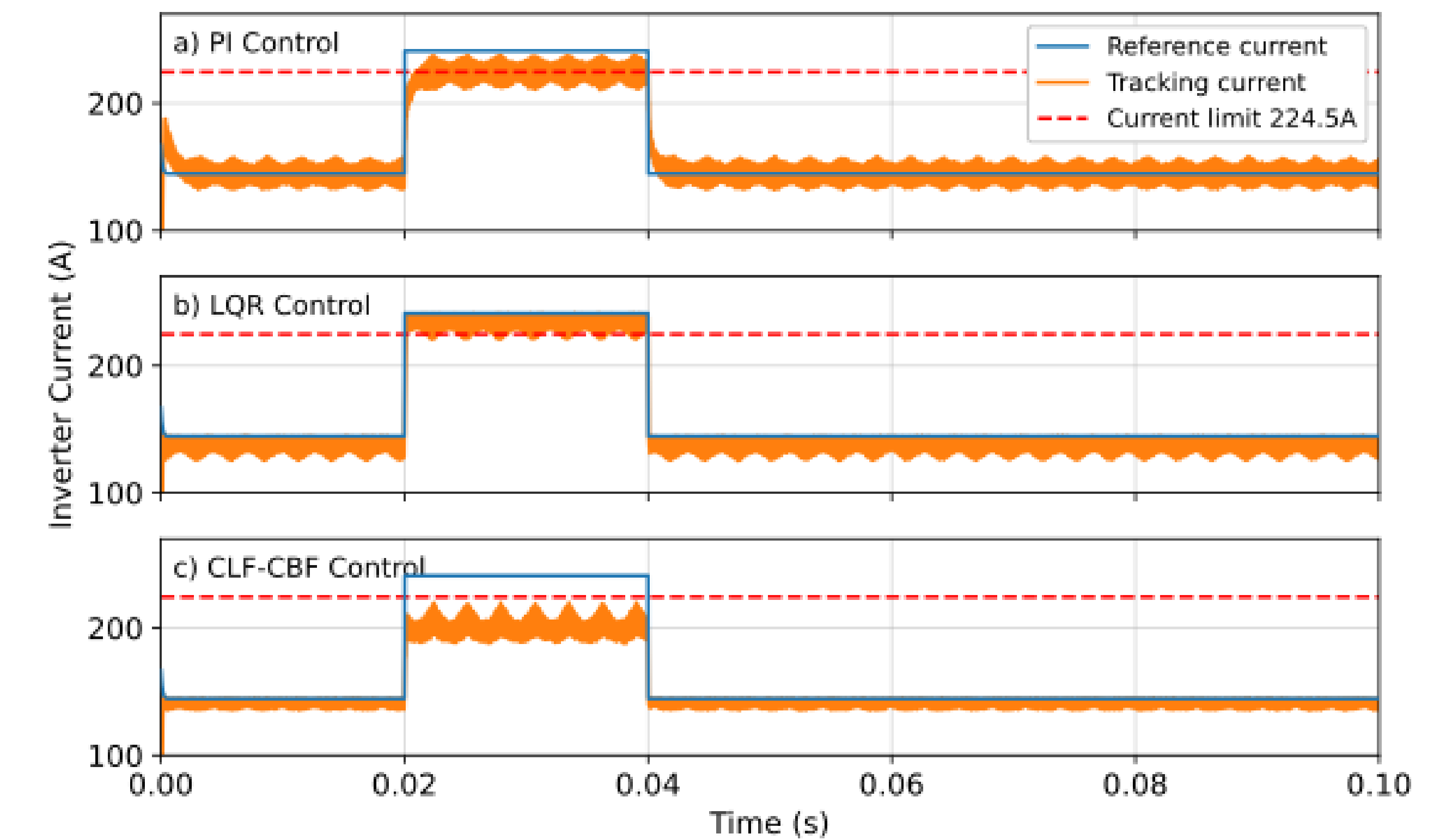


Fig. 3 – d-axis current tracking. PI overshoots and recovers slowly. LQR recovers fast but breaches the limit. CLF-CBF-QP: fast recovery, limit never crossed

- **PI** – overshoots and recovers slowly
- **LQR** – fast recovery, but breaches the limit
- **CLF-CBF-QP** – fast recovery, limit never crossed

11 Power Quality: THD

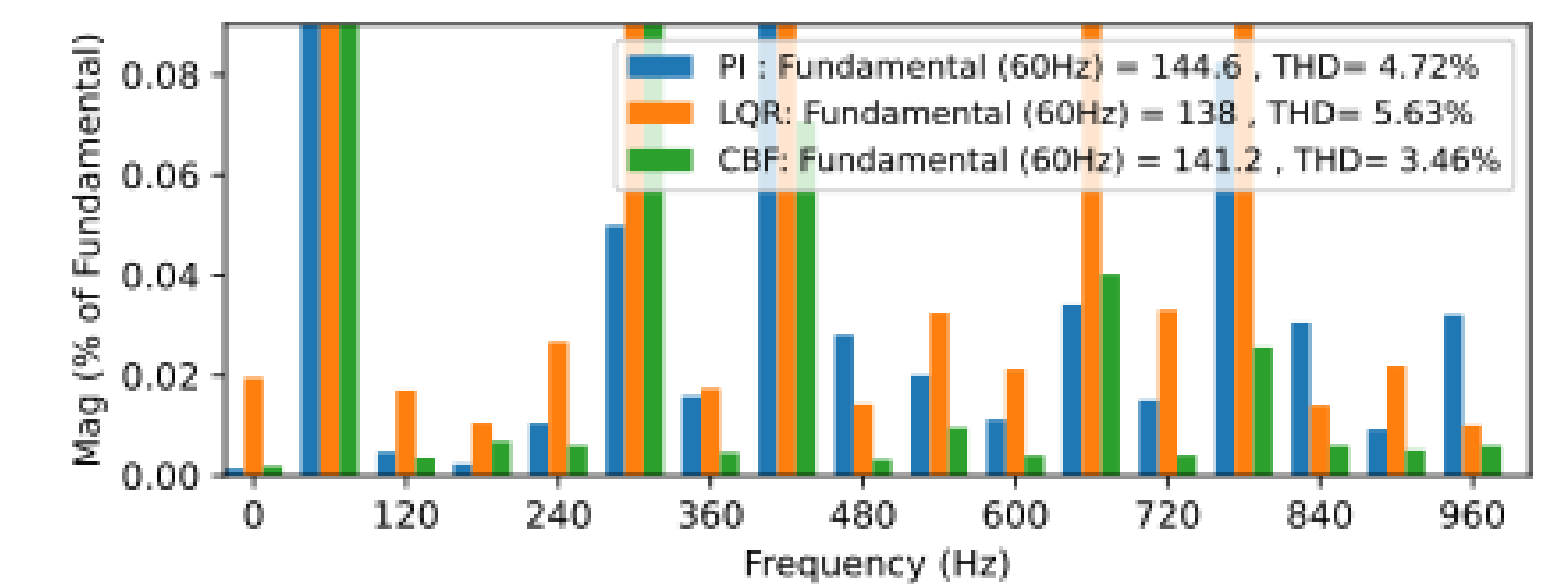


Fig. 4 – FFT of inverter output current. CLF-CBF are the smallest across the harmonic spectrum

The super-linear barrier allows smooth, minimal corrections away from the limit – less abrupt current

12 Conclusion

- **Analytical safety** – CLF-CBF embeds the current limit in the control law with formal guarantees
- **Zero violations** – CBF constraint values strictly positive throughout the voltage sag
- **Better on every metric** – lower peak current, lower THD

13 References

- [1] IEEE, “Ieee standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces,” IEEE Std 1547-2018, pp. 1–138, 2018.
- [2] B. Bahrani, S. Kennelmann, and A. Rufer, “Multivariable-pi-based dq current control of voltage source converters with superior axis decoupling capability,” IEEE TIE, vol. 58, no. 7, pp. 3016–3026, 2011.
- [3] S. A. O. D. Silva, R. A. Modesto, L. P. Sampaio, and L. B. G. Campanhol, “Dynamic improvement of a upqc system operating under grid voltage sag/swell disturbances,” IEEE Trans. Circuits and Systems II: Express Briefs, vol. 71, no. 5, pp. 2844–2848, 2024.

14 Acknowledgement

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